

Li Chuin Chong

Computational Virologist | Life Science Researcher | Bioinformatician

Ph.D. Candidate • TWINCORE GmbH c/o HZI • Hannover Medical School
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SUMMARY

I am a computational virologist working at the interface of viral evolution, large-scale sequence analysis, and data-driven modeling. My research focuses on understanding viral diversity and emergence using high-throughput sequencing data. I develop scalable computational approaches for virus discovery, diversity analysis, and spillover risk assessment, with a particular interest in quantitative and integrative frameworks for studying viral evolution.

Currently, I am a PhD candidate in Biomedical Data Science at Hannover Medical School (TWINCORE), where my work is associated with the COVIPA consortium that focus on gathering the knowledge about virological and immunological drivers of COVID-19 pathogenesis.

Virus evolution • Virus discovery • Spillover risk modelling

CURRENT APPOINTMENTS

TWINCORE Centre for Experimental and Clinical Infection Research c/o
Helmholtz Centre for Infection Research (HZI) | Hannover, DE
Institute for Experimental Virology

Doctoral Researcher

Apr 2022 - Present

EDUCATION

Hannover Medical School | Hannover Biomedical Research School, Germany

Ph.D., Biomedical Data Science

Apr 2022 - Present

- Project title: Increasing pandemic preparedness by computational high-throughput virus discovery: Identification of RNA viruses and host reservoirs with high spillover risk
- Supervisory committee: Chris Lauber (main), Stefan Seitz, Alice McHardy, Daniel P. Depledge

Perdana University | School of Data Sciences, Malaysia

MRes, Bioinformatics

2020

- Dissertation: Mapping the minimal set of the viral peptidome
- Awarded with Distinction (the highest)
- Supervisory committee: Asif M. Khan (main), Sy Bing Choi, Kenneth Ban Hon Kim

Universiti Putra Malaysia | Faculty of Medicine and Health Sciences, Malaysia

B.Sc., Biomedical Sciences

2018

- Dissertation: Expression, purification and characterisation of the protruding domain of *Macrobrachium rosenbergii* nodavirus capsid protein
- Supervisory committee: Kok Lian Ho, Abdah MD Akim

PROFESSIONAL EXPERIENCE

Doctoral Researcher

Apr 2022 - Present

Institute for Experimental Virology, TWINCORE Centre for Experimental and Clinical Infection Research c/o Helmholtz Centre for Infection Research (HZI), Germany

Visiting Trainee

Oct - Nov 2023

Division of Virus Associated Carcinogenesis (F170), German Cancer Research Center (DKFZ), Germany

Research Intern

May 2020 - Dec 2021

Faculty of Medical Technology, Mahidol University, Thailand

International Visiting Researcher

Feb - Sep 2021

Beykoz Institute of Life Sciences and Biotechnology (BILSAB), Bezmialem Vakif University, Turkey

Research Assistant

Sep - Dec 2020

Faculty of Dentistry, University of Malaya, Malaysia & Cancer Research Malaysia

Research Assistant

Dec 2019 - May 2020

Perdana University School of Data Sciences (PU-ScDS), Malaysia

SKILLS**Experimental Techniques (previously used)**

Molecular biology: Plasmid/DNA extraction, Gel electrophoresis (Agarose/SDS-PAGE), Protein expression, Western blotting (semi-dry method), Bradford assay, Protein quantification, Immobilised metal affinity chromatography (IMAC), Gravity-flow chromatography, Size exclusion chromatography (Fast protein liquid chromatography (FPLC)), Circular dichroism (CD), Dynamic light scattering (DLS), Polymerase Chain Reaction (PCR)

In vivo techniques: Rat/mice dissection, Animal phlebotomy (rat/mice/dog model)

Bioinformatics

Comparative genomics & Phylogenetics: Multiple sequence alignment, Molecular phylogenetics, Sequence clustering, Phylogenetic tree construction

Vaccine informatics & Immunoinformatics: Entropy-based quantitative analysis, Motif identification, Transmission analysis, Diversity analysis, Immune-relevance analysis

Structural bioinformatics: Protein and RNA structure annotation, Structural visualisation (PyMOL, UCSF Chimera, VMD), Secondary structure prediction, Protein primary structure analysis, Protein-protein docking analysis, Protein-protein interaction analysis

Viral informatics & Metagenomics: Viral genome annotation, SRA-based virus discovery, Metagenomic virome profiling (Taxonomic characterization of viromes, Abundance estimation of viral population), Host-pathogen interaction analysis

High performance computing (HPC) environments: PBS, SLURM

Workflow automation and management: Snakemake

Data Science

Exploratory data analysis, Data visualisation, Web scraping, Application development, Biostatistics

Web Development

HTML, CSS, JavaScript

Programming Languages (intermediate)

Python, R, Shell scripting, GitHub CI/CD pipeline

PUBLICATIONS

Citations: 395 | **h-index:** 11 | **Papers Published:** 25

For a complete list of publications, please visit my [Google Scholar profile](#).

Chong et al. Quantifying Zoonotic Spillover Risk: Identifying High Risk RNA Viruses with PREDICTORix. *Manuscript in preparation*.

(selected; *co-first author)

Chuprikova, L., **Chong, L. C.***, Ruff, S., Seitz, S., & Lauber, C. (2026). Data-driven virus discovery with virushunter and virusgatherer. *bioRxiv*. doi:[10.64898/2026.01.26.701714](https://doi.org/10.64898/2026.01.26.701714).

Chong, L. C., & Khan, A. M. (2024). A systematic bioinformatics approach for mapping the minimal set of a viral peptidome. *Current Protocols*. doi:<https://doi.org/10.1002/cpz1.1056>.

Chong, L. C., & Lauber, C. (2023). Viroid-like RNA-dependent RNA polymerase-encoding ambiviruses are abundant in complex fungi. *Frontiers in Microbiology*, 14. doi:<https://doi.org/10.3389/fmicb.2023.1144003>.

Chong, L. C., & Khan, A. M. (2022). Historical milestone in 42 years of viral sequencing — impetus for a community-driven sequencing of global priority pathogens. *Frontiers in Microbiology*, 13. doi:<https://doi.org/10.3389/fmicb.2022.1020148>.

Chong, L. C., & Khan, A. M. (2022). Negligible peptidome diversity of SARS-CoV-2 and its higher taxonomic ranks. *bioRxiv*. doi:<https://doi.org/10.1101/2022.10.31.513750>.

Chong, L. C., & Khan, A. M. (2019). Identification of serotype-specific, highly conserved dengue virus sequences: Implications for vaccine design. *BMC Genomics*, 20(Suppl 9), 921. doi:<https://doi.org/10.1186/s12864-019-6311-z>.

BOOK CHAPTERS

Chong, L. C., & Khan, A. M. (2025). Vaccine target discovery. *Encyclopedia of Bioinformatics and Computational Biology, 2nd Edition*. doi:[10.1016/B978-0-323-95502-7.00226-8](https://doi.org/10.1016/B978-0-323-95502-7.00226-8).

Chong, L. C., & Khan, A. M. (2019). Vaccine target discovery. *Encyclopedia of Bioinformatics and Computational Biology*. doi:[10.1016/B978-0-12-809633-8.20100-3](https://doi.org/10.1016/B978-0-12-809633-8.20100-3).

SOFTWARE

PREDICTORix: A scalable, phylogeny-aware pipeline for quantifying RNA virus spillover risk

 [PREDICTORix \(private mode\)](#) • [Developer \(main\)](#)

UNIQmin: An alignment-independent tool for the study of pathogen sequence diversity at any given rank of taxonomy lineage

 [PyPi](#) •  [UNIQmin](#) • [Developer \(main\)](#) and [maintainer](#)

vDiveR: A graphical user interface (GUI)-based web application hosted on R Shiny for the visualization of various diversity dynamics

 [CRAN](#) •  [vDiveR](#) • [Co-developer](#)

VirusHunterGatherer: A Snakemake pipeline for data-driven virus discovery

 [VirusHunterGatherer](#) • [Contributor](#)

Virusparies: An R package designed for visualizing outputs from VirusHunterGatherer

 [Virusparies](#) • [Contributor](#)

GRANTS

Intramural Funding

Helmholtz Visiting Researcher Grant (€4,000)

2023

Role: Visiting Trainee

Extramural Funding

PUTI Q1 Grant from Universitas Indonesia (IDR 120 million)	2022
Role: Collaborator PI: Toar Jean Maurice Lalisang (main) & Linda Erlina (co-)	
Malaysian Medical Association (MMA) (MYR 5,000)	2019
Role: Collaborator PI: Mohammad Asif Khan (main) & Choi Sy Bing (co-)	

TEACHING AND SERVICE

Graduate Student Supervisor

Supervisor

Stephen Sugiharto • B.Sc. (Bioinformatics), I3L, ID	2022-2023
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Student Mentor

Sergej Ruff • M.Sc. (Biomedical Data Science), MHH, DE	2024
Tham Hong Yun • Ph.D. (Microbiology), UPM, MY	2018-2024
Sofiah binti Tajuddin • MRes (Microbiology), UPM, MY	2019-2022
Srosha Khanjou • BSc (Medicine), MUIC, TH	2020-2021
Interns at CaRE UPM* • B.Sc. (Biomedical Sciences), UPM, MY	2017-2018
• B.Sc. (Biomedical Sciences), UM, MY	2018

*UPM: Siti Najiha Binti Abu Bakar, Anis Zuhaida Binti Mamnun, Nor Shamira Rosdi, Muhammad Rusydie Mohd Rasid, Eugene Chua, Ong Si Ying, Teo Yu Yun, Lim Xin, Khor Winnie, Lok Zen Hui, Even Lim; UM: Aisyah Shamsuddin, Wardah Othman, Nurhusna Abd Samad

Outreach & Professional Societies

Leadership

• Finance Committee Member, ISCB Student Council (ISCBSC)	2023-Present
• Organizing Committee Member, APBioNETTalks	2020-2023
• Ambassador, Asia Pacific Bioinformatics Network (APBioNET)	2021-2022
• Public Relations & Outreach (PRO) Committee Member, Global Organisation for Bioinformatics Learning, Education & Training (GOBLET)	2019-2021
• Executive Council of Association for Medical and Bio-Informatics Singapore (AMBIS)	2018-2019
• Executive Council of Alumni and Leadership of Biomedical Science Club, UPM	2015

PRESENTATIONS

Keynote/Invited talks

An overview of the RNA virus discovery field since the 2023 RdRp Summit!
11-12 May 2025

- RdRp Summit 2025, Lisbon, Portugal

Unravelling the virosphere: Insights from viral discovery and sequence diversity analysis

17 Jun 2023

- Guest Lecture series of Indonesian Medical Education and Research Institute (IMERI)

Contributed talks

PREDICTORix: A scalable phylogeny-aware approach for quantifying viral spillover risk

11-12 May 2025

- RdRp Summit 2025, Lisbon, Portugal

- A near-full compression of SARS-CoV-2 peptidome using UNIQmin 12-21 Sep 2022
- 21st European Conference on Computational Biology (ECCB2022), Stiges, Barcelona
- Identification of highly conserved, serotype-specific dengue virus sequences: Implications for vaccine design 10-12 Sep 2019
- 18th International Conference on Bioinformatics (InCoB 2019), Jakarta, Indonesia
- Posters (selected)**
- PREDICTORix: A scalable phylogeny-aware approach for quantifying viral spillover risk 13-15 May 2025
- International Virus Bioinformatics Meeting (ViBioM 2025), Lisbon, Portugal
- Towards quantifying zoonotic spillover risks of RNA viruses: A comprehensive approach using next-generation sequencing data 16-20 Sep 2024
- 23rd European Conference on Computational Biology (ECCB 2024), Turku, Finland
- Redefining the genetic diversity and host range of Flavi-like viruses 25-28 Mar 2024
- 33rd Annual Meeting of the Society for Virology (GfV2024), Vienna, Austria
- UNIQmin, an alignment-independent tool for the study of viral sequence diversity at any given rank of taxonomy lineage: HPC solutions for big-data performance boost 30 Aug-3 Sep 2021
- 2021 ACM Europe Summer School (virtual)
- Mapping the minimal set of the viral peptidome across all major viral taxonomies 25-30 Jul 2021
- 29th Conference on Intelligent Systems for Molecular Biology and 20th European Conference on Computational Biology (ISMB/ECCB 2021; virtual)
- Identification of serotype-specific, highly conserved dengue virus sequences: Implications for vaccine design 26-28 Sep 2018
- 17th International Conference on Bioinformatics (InCoB 2018), New Delhi, India

HONORS AND AWARDS

- MHH Congress Scholarship – Poster presentation, ViBioM 2025
- ISME Travel Fellowship – Oral presentation, RdRp Summit 2025
- Best Project Award, Collaborative Bioinformatics Hackathon 2025: Machine learning and AI approaches to multimodal problems in computational biology
- OBF Travel Fellowship – Poster presentation, ECCB 2024
- GOBLET Fund & Malaysia Trust Fund Travel Fellowship – Oral presentation, ECCB 2022
- First Prize, BCM/HGSC Virtual Hackathon 2021: Pan-structural Variation in the Cloud
- ESCV Fellowship – Poster presentation, ESCV 2021
- ISCB Fellowship – Poster presentation, ISMB/ECCB 2021
- Malaysia Trust Fund – Support for publication fee Chong et al. (2021)
- Best E-Poster Award, ICBCBBS 2021
- APBioNET Travel Fellowship – Oral presentation, InCoB 2019
- ISCB Travel Fellowship – Poster presentation, InCoB 2018
- Public Service Department - National Degree Programme (JPA-PIDN) Scholarship for B.Sc. (Biomedical Sciences), 2015-2018

REFERENCES

Dr. Chris Lauber

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Institute of Experimental Virology,
TWINCORE GmbH
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RESIST Cluster of Excellence, Medizinische
Hochschule Hannover (MHH)
Hannover, Germany
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Institute of Experimental Virology,
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